

General Instructions for the candidate

- 1. This test paper contains 100 multiple choice questions. Each question has 4 alternative choices marked as a, b, c and d. The candidate must choose the most appropriate option out of the four alternatives. The candidate must attempt all the questions.**
- 2. For each correct answer, four marks shall be awarded and one mark shall be deducted for each incorrect answer. Zero marks will be awarded for unanswered question.**
- 3. The test is of two and half-hour duration (150 minutes) and carries 400 marks.**

- Q1. A man holds a book weighing 5 N between hands and keeps it free from falling by pressing both hands against the book with a force of 10 N each. The coefficient of friction between the book and hand would be to _____.
 a. 0.15
b. 0.25
 c. 0.45
 d. 0.8
- Q2. The magnitude of shear stress induced in a solid shaft due to applied torque varies _____.
 a. From maximum at the centre to zero at the circumference.
b. From zero at the centre to maximum at the circumference.
 c. From maximum at the centre to minimum at the circumference and not zero.
 d. From minimum at the centre and not zero to maximum at the circumference.
- Q3. A hollow shaft of the same cross-sectional area as that of a solid shaft can _____.
 a. resists less torque
b. resists more torque
 c. resists equal torque
 d. none of the above
- Q4. The equivalent length of a column fixed at both the ends is _____.
 a. 0.7 L
b. 0.5 L
 c. L
 d. 2 L
- Q5. The principal stresses σ_1 , σ_2 and σ_3 at a point respectively are 80 MPa, 30 MPa and -40 MPa. The maximum shear stress is _____.
 a. 25 MPa
 b. 35 MPa
 c. 55 MPa
d. 60 MPa
- Q6. Which one of the following is preferable cross-section of a beam for bending loads?
 a. Circular
 b. Annular circular
 c. Rectangular
d. I-section

- Q7. A particle starts from rest and moves in a straight line whose equation of motion is given by $S = 2t^3 - t^2 - 1$. The acceleration of the particle after three seconds will be _____.
- 40 m/s²
 - 60 m/s²
 - 28 m/s²
 - 34 m/s²
- Q8. A rubber ball is thrown vertically upward with a velocity u from the top of a building. It strikes the ground with a velocity $3u$. The time taken by the ball to reach the ground is given by _____.
- 4u/g
 - 3u/g
 - 2u/g
 - u/g
- During inelastic collision of two particles, which one of the following is conserved?
- total linear momentum only
 - total kinetic energy only
 - both linear momentum and kinetic energy
 - neither linear momentum nor kinetic energy
- Q10. The moment of inertia of a triangular section of base 'a' and height 'L' about an axis passing through its CG and parallel to the base is _____.
- $aL^3/36$
 - $aL^3/6$
 - $aL^3/8$
 - $aL^3/12$
- Q11. The forces which meet at one point and have their line of action in different planes are called _____.
- Intersecting forces
 - Non-coplanar non-concurrent forces
 - Coplanar non-concurrent forces
 - Non-coplanar concurrent forces
- Q12. What is the magnitude of the resultant of two like parallel forces 10 N and 30 N separated by a distance of 10 cm?
- $20\sqrt{2}$ N
 - $10\sqrt{10}$ N
 - 20 N
 - 40 N

- Q13. Which one of the following forces is a contact force?
- Force of gravity
 - Magnetic force
 - Force of friction
 - Electrostatic force
- Q14. Which of the following is CORRECT for indeterminate beam condition?
- Number of unknown components should be equal to the number of equilibrium equations
 - Number of unknown components should be less than the number of equilibrium equations
 - Number of unknown components should be greater than the number of equilibrium equations
 - Number of unknown components should be zero
- Q15. A triangular shaped frame is a/an _____ frame.
- redundant
 - imperfect
 - deficient
 - perfect
- Q16. A block of mass 5 kg slides down from rest along a frictionless inclined plane that makes an angle of 30° with horizontal. What will be the speed of the block after it covers a distance of 3.6 m along the plane? [$g = 10 \text{ m/s}^2$].
- 5 m/s
 - 6 m/s
 - 7 m/s
 - 8 m/s
- Q17. A thin disc and a thin ring, both have mass M and radius R . Both rotate about axes through their centre of mass and are perpendicular to their surfaces at the same angular velocity. Which of the following is true?
- The ring has higher kinetic energy
 - The disc has higher kinetic energy
 - The ring and the disc have the same kinetic energy
 - Kinetic energies of both the bodies are zero since they are not in linear motion
- Q18. The free body diagram of a body shows the body _____.
- with its surroundings and external forces acting on it
 - isolated from all external forces
 - isolated from all surroundings
 - isolated from its surroundings and all external effects acting on it

Q19. Which of the following produces least friction?

- a. Sliding friction
- ☒ b. Rolling friction
- c. Static friction
- d. Same in static and dynamic friction

Q20. A book of mass 10 kg is lying at the front desk of a classroom. The coefficient of static friction is 0.5. A force of 40 N is applied to the book. The book will _____.



- a. move
- ☒ b. will not move
- c. cannot say
- d. none of the above

Q21. A body of weight 'W' is required to move up on rough inclined plane whose angle of inclination with the horizontal is ' α '. The effort applied parallel to the plane is given by (where $\mu = \tan \phi$ = Coefficient of friction between the plane and the body.)

- a. $P = W \tan \alpha$
- ☒ b. $P = W \tan (\alpha + \phi)$
- c. $P = W (\sin \alpha + \mu \cos \alpha)$
- d. $P = W (\cos \alpha + \mu \sin \alpha)$

Q22. The sum of magnitudes of two forces acting at a point is 16 N. If the resultant force is 8N and its direction is perpendicular to smaller force, then the forces are _____.

- ☒ a. 6N and 10N
- b. 8N and 10N
- c. 6N and 12N
- d. 10N and 10N

Q23. Maximum deflection for a cantilever beam of span L, loaded at free end by force P is given by _____.

- a. $PL^2/3EI$
- b. $PL^2/6EI$
- c. $PL^3/8EI$
- ☒ d. $PL^3/3EI$

4. Calculate Young's Modulus of Elasticity of a material, whose bulk modulus is 50 GPa, modulus of rigidity is 30 GPa and Poisson's ratio is 0.25.

- a. 75 GPa

- b. 110 GPa
- c. 60 GPa
- d. 90 GPa

Q25. Compared with solid section, hollow section column has _____.

- a. High buckling strength
- b. Low buckling strength
- c. Same buckling strength
- d. None of the above

Q26. Which of the following is false?

- a. $f(a)$ is an extreme value of $f(x)$ if $f'(a) = 0$
- b. If $f(a)$ is an extreme value of $f(x)$, then $f'(a) = 0$
- c. If $f'(a) = 0$, then $f(a)$ is an extreme value of $f(x)$
- d. None of these

Q27. For what value of λ , do the simultaneous equation $4x + 3y = 1$, $8x + 6y = \lambda$ have infinite solutions?

- a. $\lambda = 0$
- b. $\lambda = 1$
- c. $\lambda \neq 2$
- d. $\lambda = 2$

Q28. Stokes theorem connects _____.

- a. a line integral and a surface integral
- b. a surface integral and a volume integral
- c. a line integral and a volume integral
- d. gradient of a function and its surface integral

Q29. The family $y = Bx + B^4$ of curves is represented by differential equation of _____ degree.

- a. 1
- b. 2
- c. 3
- d. 4

Q30. What is the number of ways that a team of 11 players can be made out of 14 players?

- a. 364
- b. 1001
- c. 1365
- d. More than one of the above

Q31. The three vertices of a parallelogram ABCD are A (1, 2, 3), B (-1, -2, -1) and C (2, 3, 2). The coordinates of the fourth vertex D are _____.

- a. (-2, -3, 0)
- b. (4, 7, 6)
- c. (2, 3, 4)
- d. (2, 2, 3)

Q32. The curve for which the slope of the tangent at any point is equal to the ratio of the abscissa to the ordinate of the point is _____.

- a. ellipse
- b. parabola
- c. circle
- d. rectangular hyperbola

Q33. The point with position vectors $5\hat{i} - 2\hat{j}$, $8\hat{i} - 3\hat{j}$ and $a\hat{i} - 15\hat{j}$ are collinear if the value of a is _____.

- a. 31
- b. 41
- c. 42
- d. 44

Q34. Multiplication of real valued square matrices of same dimension is _____.

- a. Associative
- b. Commutative
- c. Always positive definite
- d. not always possible to commute

Q35. Value of $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{x+2}$ is equal to _____.

- a. e
- b. e^{-1}
- c. e^{-2}
- d. None of these

Q36. The number of ways in which 5 white balls and 3 red balls are to be placed around a circle so that no two white balls are placed together:

- a. 1460
- b. 1440
- c. 1420
- d. 1410

Q37. If $u = (x^2 + y^2 + z^2)^{1/2}$, then $\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$ is equal to _____.

- a. 4u
- b. $2/u$
- c. 2u

d. $-u/4$

Q38. The value of the integral $\int_{-2}^2 |1 - x^2| dx$ is _____.

☒ a. 0

b. 4

c. 1

d. ∞

Q39. The integrating factor of differential equation $\frac{dy}{dx} + y \tan x - \sec x = 0$ is _____.

a. $\cos x$

☒ b. $\sec x$

c. $e^{\cos x}$

d. $e^{\sec x}$

Q40. The modulus of eigenvalues of an orthogonal matrix is _____.

a. 0

b. -1

c. 1

☒ d. 2

Q41. Which one of the following statements is true for all real symmetric matrices?

☒ a. All the eigenvalues are real.

b. All the eigenvalues are positive.

c. All the eigenvalues are distinct.

d. Sum of all the eigenvalues is zero

Q42. The value of $\lim_{x \rightarrow \infty} 2x \sin\left(\frac{4}{x}\right)$ is _____.

a. 2

b. 4

☒ c. 8

d. 1

Q43. When the height of a building is equal to the length of its shadow, what is the angle of elevation of the Sun?

a. 30°

☒ b. 45°

c. 90°

d. 60°

Q44. Find the value of $\sin(1920^\circ)$?

a. $1/2$

b. $1/\sqrt{2}$

- c. $\sqrt{3}/2$
d. $1/3$

Q45. If $3x^2 + ax + 4$ is perfectly divisible by $x - 5$, then the value of a is _____.

- a. -12
b. -5
c. -15.8
d. -14.5

Q46. If we add two irrational numbers the resulting number _____.

- a. is always a rational number
b. is always an irrational number
c. may be a rational or an irrational number
d. always an integer

Q47. The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}$ is _____.

- a. 0
b. $-\pi/4$
c. $\pi/2$
d. $\pi/4$

Q48. If the order of A is 4×3 , the order of B is 4×5 and the order of C is 7×3 , then the order of $(A^T B)^T C^T$ is _____.

- a. 5×3
b. 4×5
c. 5×7
d. 4×3

Q49. What is the degree of the equation $\frac{1}{x-3} = \frac{1}{x+2} - \frac{1}{2}$?

- a. 0
b. 1
c. 2
d. 3

Q50. If the difference between the mode and median is 4, then find the difference between the median and mean (in the given order).

- a. 2
b. 1
c. 3
d. 4

Q51. Which function key is used for opening help?

- a. F1

- b. F2
- c. F4
- d. F3

Q52. What is the full form of PNG in the computer image format?

- a. Printable New Graphic
- b. Printable New Graphical
- c. Port Natural Graphics
- ☒ d. Portable Network Graphics

Q53. Which of the following belongs to the category of fourth generation languages?

- a. Machine language
- ☒ b. High-level language
- c. Assembly language
- d. Query language

Q54. In a flow chart, the process box is represented by _____.

- ☒ a. Rectangular
- b. Diamond
- c. Parallelogram
- d. Oval

Q55. Representation of steps to solve a problem with the help of a diagram is known as:

- a. Graph
- b. Program
- c. Algorithm
- ☒ d. Flowchart

Q56. The sequence of steps written independent of a specific programming language to solve a problem.

- a. Coding
- b. Processing
- ☒ c. Algorithm
- d. None of the above

Q57. Which of the following is the shortcut key to open a 'New Blank' document in MS-Word?

- a. CTRL + B
- ☒ b. CTRL + N
- c. CTRL + D
- d. CTRL + M

Q58. Light year is a unit of _____.

- a. Time

- b. Distance
c. Light
d. Intensity of light
- Q59. How much work is done in moving a charge of 5C across two points having a potential difference of 16 V?
a. 65 J
b. 45 J
c. 40 J
d. 80 J
- Q60. A car A is moving with a speed of 40 km/h. Another car B is moving with 60 km/h in the opposite direction to car A. What will be the relative speed of car B with respect to car A?
a. 40 km/h
b. 60 km/h
c. 100 km/h
d. 20 km/h
- Q61. The area under the velocity-time graph gives _____.
a. acceleration
b. total distance travelled
c. speed
d. retardation
- Q62. A sphere P impinges directly onto another identical sphere Q at rest. If the coefficient of restitution is 0.5, the ratio of velocities V_q / V_p after the impact would be _____.
a. 1 : 1
b. 2 : 1
c. 3 : 1
d. 2 : 3
- Q63. Strain in a given material is the ratio of _____.
a. applied load to the area of cross-section opposing the load
b. change in dimension to the original dimension
c. applied load to the original dimension
d. change in dimension to the area of cross-section opposing the load
- Q64. A man walked 60 meter to the East from point A, then turn right and walked 50 meter. In last he turn left and went for 60 meter and reached at point B. Find AB?
a. 170 meter
b. 130 meter
c. 145 meter

d. 150 meter

Q65. A cubical block rests on an inclined plane of $\mu = \frac{1}{\sqrt{3}}$, determine the angle of inclination when the block just slides down the inclined plane?

- a. 40°
- b. 50°
- c. 60°
- d. 20°

Q66. The water droplets in free fall are spherical due to _____.

- a. Gravity
- b. Viscosity
- c. Surface tension
- d. Intermolecular attraction

Q67. A position dependent force, $F = 7 - 2x + 3x^2$ N acts on a small body of mass 2kg and displaces it from $x = 0$ to $x = 5$ m. The work done in joules is _____.

- a. 135
- b. 270
- c. 35
- d. 70

Q68. Chemical name of washing soda is _____.

- a. Sodium chloride
- b. Sodium hydrogen carbonate
- c. Sodium carbonate
- d. Sodium hydroxide

Q69. Which is the most abundant metal in the human body?

- a. Zinc
- b. Iron
- c. Magnesium
- d. Calcium

Q70. Which one of the following is not a colloid?

- a. Aqueous solution of NaCl
- b. Milk
- c. Water
- d. Iron

Q71. At which point of the Earth is there no gravity?

- a. At North and South Pole
- b. At equator
- c. On the ocean surface
- d. At centre of the Earth

- Q72. Which force moves an aircraft through the air?
- Lift Force
 - ☒ Drag Force
 - Thrust Force
 - Gravity Force
- Q73. Why are the soles of the shoes treaded?
- To give the shoes more protection
 - To decrease friction
 - ☒ To increase friction
 - To increase life of the shoes
- Q74. Pointing to a man in a photograph a man said to a woman “his mother is the only daughter of your father”. How is the woman related to the man in the photograph?
- Sister
 - ☒ Mother
 - Wife
 - Daughter
- Q75. In a certain code language, ‘kew xas huma deko’ means ‘she is eating apples’, ‘kew tepo qua’ means ‘she sells toys’ and ‘su lim deko’ means ‘I like apples’. Which word in that language means she and apples?
- Xas & deko
 - ☒ Kew & deko
 - Xas & kew
 - Kew & xas
- Q76. Look at this series: 2, 1, (1/2), (1/4), ... What number should come next?
- (1/3)
 - ☒ (1/8)
 - (2/8)
 - (1/16)
- Q77. Marks is three times older than Jacks. 2 years ago, he was five times older than Jacks. What is the present age of Marks?
- ☒ 12 years
 - 14 years
 - 18 years
 - 24 years
- Q78. Radha keeps all his socks in a single drawer. She has 24 pairs of white socks and 18 pairs of grey socks. Radha picks 3 socks randomly. Find the possibility of Radha choosing a matching pair.

- a. $1/36$
- b. $1/108$
- c. $7/36$
- ☒ d. $1/1$

Q79. A toy was sold at a profit of 20% on the cost price. If the toy was sold at 10% lower than this price, then the profit would have been Rs. 600 less. At what price should the toy be sold if one wants to gain 25% of the cost price?

- a. 7500
- b. 5000
- c. 6250
- d. 7250

Q80. Statement: It is desirable to put the child in school at the age of 5 or so.

Assumptions:

- I. At that age the child reaches appropriate level of development and is ready to learn.
 - II. The schools do not admit children after six years of age.
- ☒ a. Only assumption I is implicit
 - b. Only assumption II is implicit
 - c. Either I or II is implicit
 - d. Neither I nor II is implicit

Q81. Fill the blank for the given series

SCD, TEF, UGH, _____, WKL

- a. CMN
- b. UJI
- ☒ c. VIJ
- d. IJT

Q82. What is the number of triangles that can be formed whose vertices are the vertices of an octagon but have only one side common with that of octagon?

- a. 64
- ☒ b. 32
- c. 24
- d. 16

Q83. What will come in place of the question mark (?) in the following number series?
5, 7, ?, 25, 45, 75

- a. 11
- ☒ b. 13
- c. 15
- d. 19

Q84. Complete the analogous pair

Vigilant : Alert : : Viable : ?

- a. Active
- b. Hopeless
- ☒ c. Feasible
- d. Useful

Q85. What will come in place of the question mark (?)

121, 222, 424, ?

- a. 646
- ☒ b. 828
- c. 626
- d. 524

Q86. Traffic on a highway is moving at a rate of 360 vehicles per hour at a location. If the number of vehicles arriving on this highway follows Poisson distribution, the probability (round off to 2 decimal places) that the headway between successive vehicles lies between 6 and 10 seconds is _____.

- ☒ a. 0.18
- b. 0.37
- c. 0.85
- d. 0.97

Q87. If each side of a square is increased by 25%, find the percentage change in its area?

- a. 65.25
- ☒ b. 56.25
- c. 65
- d. 56

Q88. Four bells ring simultaneously at starting and an interval of 6 sec, 12 sec, 15 sec and 20 sec respectively. How many times they ring together in 2 hours?

- a. 120
- b. 60
- ☒ c. 121
- d. 112

Q89. Given below a statement followed by 2 inferences. Choose the inference that can be inferred from the given statement by selecting the right option.

Statement: Out of 20 students in a class, 15 passed in English and 18 passed in mathematics.

Inferences:

- I. There are few students who passed in both the subjects
- II. No student failed in both the subjects

- a. Both inferences I and II follow

- ☒ Only inference I follows
☐ Only inference II follows
☐ Neither inference I nor inference II follows
- Q90. In a row of boys, A is 13th from the left and D is 17th from the right. If in this row A is 11th from the right, then what is the position of D from the left?
- ☐ 6th
☒ 7th
☐ 10th
☐ 12th
- Q91. In a certain code language, the word "PENCIL" is coded as "CEILNP". How is the word "MOBILE" coded in that language?
- ☒ OBELIM
☐ BEILMO
☐ LOBMEI
☐ BOMLEI
- Q92. How many meaningful four-letter English words can be formed from the second, third, fifth, and ninth letters from the left end of the word 'TROPHOGENIST'?
- ☐ Two
☐ None
☐ Three
☒ One
- Q93. If elephant is called 'dog', 'dog' is called 'crow', 'crow' is called 'tiger', 'tiger' is called 'snake', 'snake' is called 'lion', which is reared as pet?
- ☒ Crow
☐ Tiger
☐ Snake
☐ Elephant
- Q94. Six persons A, B, C, D, E and F are sitting in two rows, three in each. E is not at the end of any row. D is second to the left of F. Also C, the neighbour of E, is sitting diagonally opposite to D. B is the neighbor of F. Which of the following are sitting diagonally opposite to each other?
- ☐ F and C
☒ D and A
☐ A and C
☐ A and F
- Q95. Given below are 2 statements followed by 2 conclusions. Choose the conclusion/conclusions that follow(s) the given statements by selecting the right option.

Statements:

I. Some bottles are bags.

II. All bags are pencils.

Conclusions:

I. No bottle is a pencil.

II. Some bottles are pencils.

- a. Both the conclusions I and II follow
- b. Neither conclusion I nor conclusion II follows
- c. Only conclusion I follows
- ☒ d. Only conclusion II follows

Q96. Which is the next number in the series given below?

12, 98, 24, 49, 48, 24.5, ?

- ☒ a. 96
- b. 86
- c. 24
- d. 49

Q97. Read the given information and answer the question asked by choosing the most appropriate option. Among 200 people, 90 people like tea while 108 people like coffee and 46 people like both tea and coffee. How many people like neither tea nor coffee?

- a. 46
- b. 44
- c. 62
- ☒ d. 48

Q98. Frog: Aestivation In a class of 120 students, 40% are females and rest are males. 50% of the female students passed an entrance exam. If the total number of students who passed the exam is 72, then what percentage of the male students passed the exam?

- a. 33.33%
- ☒ b. 66.66%
- c. 16.66%
- d. 52.25%

Q99. In three of the options, the 2nd number is related to the 1st number in a similar logical way. Which is the odd one out?

- a. 36 - 9
- b. 24 - 6
- ☒ c. 42 - 8
- d. 27 - 9

Q100. In the following question, select the related word pair from the given alternatives.

Bear: Hibernation :: ? : ?

- a. Man : Immigration
- b. Bird : Migration
- c. Food : Adulteration
- d. Frog : Aestivation

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